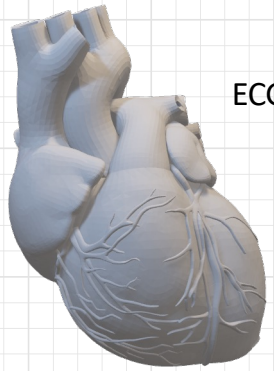
Lab 3: Mini bio-amplifer project

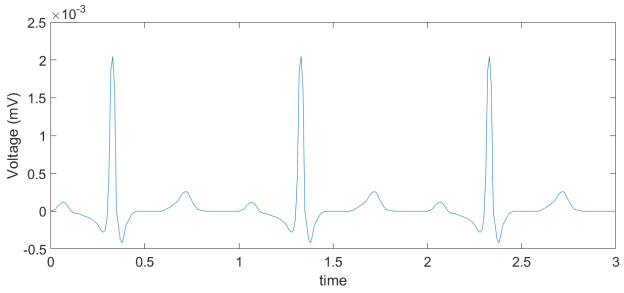
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Task 12: Recording an ECG signal.

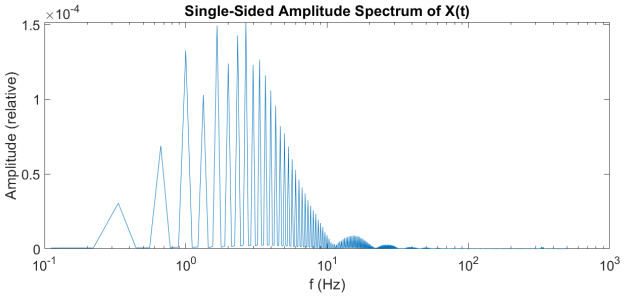
An electrocardiogram is a standard way of recording cardiac activity from a point in the surface, usually the chest. The activity profile looks like the below.



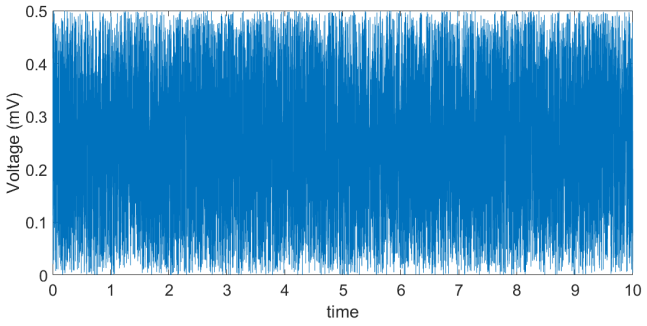
ECG Recording

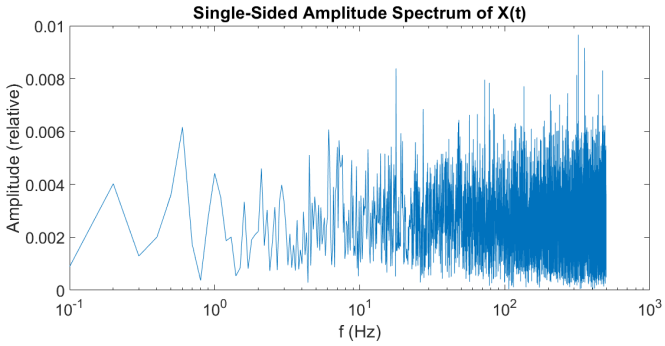


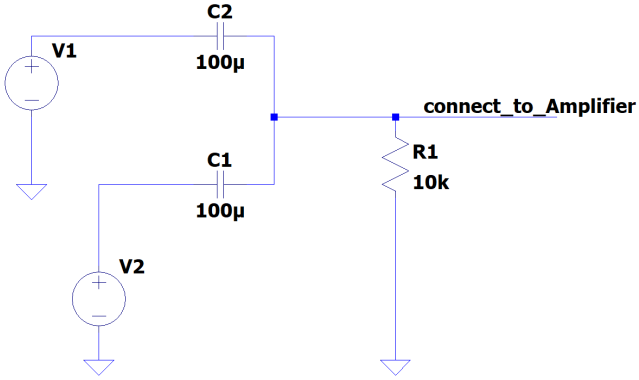
We can use an FFT to understand the frequencies used in the information.



In real world recordings, there will also be noise. So we will use a noise source as described below.







We can simulate this using the circuit to the left:
V1 can represent the ECG data. Use PWL file “ECG_PWL_Data” to create the ECG waveform
V2 can represent the noise. Use PWL file “noise_PWL” to create the noise

Use a 3s transient simulation to check each of the waveforms and the combined one.

Design an amplifier that can increase the signal range between to between 0 – 1.8V to match the requirements for an ADC. Filter the signal (low pass and high pass) to remove the noise and ensure anti aliasing. The ADC samples at 1ksamples/s.

Lab 3: Mini bio-amplifer project

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Step 1: Analyse the ECG Signal amplitude:

(i) Do a transient simulation of the ECG signal over 5seconds
(ii) What is the amplitude of the signal?
(iii) How much larger (gain) does it need to be?

Step 2: Analyse the ECG Signal range:

(i) Do a transient simulation of the ECG signal over 5seconds as per above
(ii) Right clock on your graph and choose View -> FFT. Then choose the waveform you have (it usually defaults to the one you just clicked).
(iii) In the FFT graph, right click on the y-axis and change to “linear”. Then right click on the x axis and choose a range between 100mHz and 1kHz.
(iv) Now, look at the graph and decide what frequency range the signal exists in.

Step 3: Design your amplifier!

When you are done, you need to show you demonstrator:

(i) Your circuit!
(ii) A 5ms time domain signal – to show that your amplifier hasn’t distorted the signal
(iii) A frequency analysis – to show the bandwidth of your amplifier system. To this, you will need to disconnect the ECG/Noise input and replace it with an ac voltage input.